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United States Patent	12416375
Kind Code	B2
Date of Patent	September 16, 2025
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End-piece of a flexible pipe for transporting fluid, associated flexible pipe and method

Abstract

An end-fitting of a flexible pipe for transporting fluid, that includes at least one sealing assembly with at least one crimping ring comprising at least one bulge inserted radially into a polymer sheath of the flexible pipe, the end-fitting including an annular seal, disposed in contact with the polymer sheath at a distance from the crimping ring, the annular seal having an annular central region that is deformable radially towards the polymer sheath, and two stiffened annular peripheral regions that are situated on either side of the central region along a central axis of the flexible pipe, the peripheral annular regions being movable towards one other to activate radial deformation of the central region towards the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.

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Appl. No.:	18/705524
Filed (or PCT Filed):	October 27, 2022
PCT No.:	PCT/EP2022/080119
PCT Pub. No.:	WO2023/073122
PCT Pub. Date:	May 04, 2023

Prior Publication Data

Foreign Application Priority Data

FR21 11543Oct. 29, 2021

Publication Classification

Int. Cl.: F16L33/01 (20060101); F16L11/08 (20060101); F16L33/00 (20060101)

U.S. Cl.:

CPC F16L33/01 (20130101); F16L11/083 (20130101); F16L33/003 (20130101);

Field of Classification Search

CPC: F16L (33/01); F16L (11/083); F16L (33/003); F16L (33/222); F16L (33/28)

USPC: 285/341; 285/339; 285/222.2; 285/222.4; 285/222.1; 285/255

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a 35 U.S.C. §§ 371 national stage application of International Application No. PCT/EP2022/080119, filed Oct. 27, 2022, which claims priority to French Patent Application FR2111543, filed Oct. 29, 2021, the contents of which are incorporated herein by reference.

FIELD

(2) The present invention relates to an end-fitting of flexible pipe for transporting fluid, including: at least one end region of a polymer sheath of the flexible pipe; end sections of armor elements of the flexible pipe arranged around or in the end region of the polymer sheath; an end vault defining a central bore for letting the fluid flow, extending along a central axis; a cover defining, along with the end vault, a chamber for receiving the end sections of armor elements; at least one sealing assembly comprising at least one crimping ring, the crimping ring comprising at least one bulge inserted radially into the polymer sheath; an annular seal, arranged in contact with the polymer sheath away from the crimping ring.

BACKGROUND

(3) The pipe is more particularly an unbonded flexible pipe intended for transporting hydrocarbons through a body of water, such as an ocean, a sea, a lake or a river.

(4) Such a flexible pipe is e.g. produced according to the normative documents API 17J, 4^{sup}.th edition May 2014 and API RP 17B, 5^{sup}.th edition—May 2014 established by the American Petroleum Institute.

(5) The pipe is usually formed by a set of concentric and superimposed layers. The pipe is considered to be “unbonded” as defined by the present invention provided that at least one of the layers of the pipe is configured to move longitudinally with respect to the adjacent layers during a bending of the pipe. More particularly, an unbonded pipe is a pipe which has no bonding materials connecting the layers forming the pipe.

- (6) The pipe is generally arranged across a body of water, between a bottom unit intended for collecting the fluid extracted at the bottom of the body of water and a floating surface unit intended for collecting and distributing the fluid. The surface unit can be a semi-submersible platform, an FPSO or another floating unit.
- (7) In a known manner, the pipe includes a pressure sheath made of polymer, intended to contain the fluid transported by the pipe, an intermediate sheath, if appropriate, and an external sheath intended to protect the pipe externally.
- (8) The ends of the pipe have end-fittings for connection to the bottom unit and to the surface unit.
- (9) The end region of each sheath is received in an end-fitting of the pipe and is mounted in the end-fitting in so as to be sealed.
- (10) To this end, a known solution for making a sealed connection is to crimp a metal ring around the outer face of the sheath so that the ring partially penetrates into the thickness of the sheath wall.
- (11) The crimped ring is generally in the form of a hollow cone surrounding the sheath, the cone being arranged coaxially with the sheath. During the crimping, a conical bearing surface machined in the body of the end-fitting is thrust axially around the cone, the conical seat also being coaxial with the sheath to be crimped and with the cone.
- (12) During the thrust, the cone is compressed radially with respect to the axis of the sheath, the diameter thereof decreases and the cone deforms plastically by penetrating partially inside the sheath. The crimping operation simultaneously provides the metal/polymer seal between the cone and the sheath and the metal/metal seal between the cone and the body of the end-fitting.
- (13) Such solution withstands very high internal pressures (more than 1000 bar) within a wide temperature range (up to 130° C. and above). The solution is particularly effective when the pressure inside the sheath is greater than the pressure outside the sheath, because the difference in pressure tends to increase the diameter of the sheath. As a result, the contact pressure increases between the sheath and the crimping ring, and the pressure between the crimping ring and the body of the end-fitting increases as well, thus reinforcing the sealing of the assembly.
- (14) In the case of a pressure sheath, the internal pressure is usually much higher than the external pressure, which reinforces the tightness, but the pressure gradient can sometimes be reversed, e.g. during a production shutdown followed by a decompression.
- (15) When the pressure gradient is reversed, the opposite effect to the effect described hereinabove occurs. The contact pressures between the crimping ring and the sheath are reduced and the sealing of the assembly is decreased. In particular cases, micro-leaks can be generated therefrom.
- (16) In the case of the outer sheath, the problem is more critical because the pressure gradient is often reversed permanently, in particular when the end-fitting is immersed at great depth. In such case, the hydrostatic pressure applied against the outer face of the outer sheath is significantly higher than the pressure prevailing in the wall of the pipe which applies against the inner face of the outer sheath.
- (17) To overcome such problems, it is known, e.g. from WO2019/137591, to add a sealing ring in addition to the polymer crimped ring between the end-fitting vault and the sheath the sealing of which is to be achieved.
- (18) In certain cases, the annular seal may include an external radial reinforcement region located radially on the outside of an elastomer ring.
- (19) Such a solution is not entirely satisfactory. In certain circumstances, the contact pressure applied to the polymer sheath is not strong enough to maintain the sealing. Furthermore, the solution presented in the aforementioned document applies only to the pressure sheath and is not necessarily suited for intermediate sheaths or external sheaths that may be present within the flexible pipe.

SUMMARY

- (20) A goal of the invention is thus to provide an end-fitting of flexible pipe for transporting a fluid, wherein the sealing between a polymer sheath and the vault of the end-fitting is improved,

whatever the differential pressure conditions applying to the pipe, and regardless of the type of sheath around which the seal is made in the end-fitting.

(21) To this end, the subject matter of the invention is an end-fitting of flexible pipe for transporting a fluid of the aforementioned type, characterized in that the annular seal includes an annular central region deformable radially toward the polymer sheath, and two stiffened annular peripheral regions, located on both sides of the central region along the central axis, the peripheral annular regions being movable toward each other to activate radial deformation of the central region toward the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.

(22) The end-fitting according to the invention can comprise one or a plurality of the following features, taken individually or according to any technically possible combination: at least in the active position, the central region defines a radial tip protruding toward the polymer sheath beyond each peripheral annular region; in the rest configuration, the central region has the radial tip; the central region is made of elastomer, in particular a fluoroelastomer such as FKM, FFKM or FVMQ, or a nitrile elastomer such as NBR or HNBR, each peripheral region being made of metal, or of a polymer which is more rigid than the elastomer, in particular containing PEK, PEEK, PEEKK, PEKEKK, or containing a fluoropolymer, such as polyethylene tetrafluoride (PTFE); the end-fitting defines a housing for receiving the annular seal, the housing being delimited by a transverse activation surface of the annular seal, located on the end vault or on a part fastened with respect to the end vault, the transverse activation surface extending opposite the first peripheral region, the housing being also delimited by a transverse thrust surface of the annular seal, located opposite the transverse activation surface, on an additional compression activation part of the annular seal, the transverse thrust surface extending opposite the second peripheral region, the activation transverse surface and the thrust transverse surface being movable relative to each other at least during the fitting of the end-fitting, so as to switch the annular seal from the rest configuration thereof to the active configuration thereof; the additional part is fastened on the crimping ring; the end-fitting defines a pressure transmission channel external to the flexible pipe, opening out into the housing between the activation surface and the first peripheral region; the first peripheral region and/or the second peripheral region are assembled onto surfaces of the central region; the central region has a first surface inclined at a non-zero angle with respect to a plane perpendicular to the central axis, the first peripheral region having a first supplementary inclined surface directly mounted on the first inclined surface, the inclination of the first inclined surface being directed toward the second peripheral region when moving radially toward the central axis on the first inclined surface; the central region has a second inclined surface at a non-zero angle with respect to a plane perpendicular to the central axis, the second inclined surface being located opposite the first inclined surface along the central axis, the second peripheral region having a supplementary second inclined surface fastened to the second inclined surface, the central region advantageously having a trapezoidal cross-section; the first peripheral region and the second peripheral region each have a continuous or discontinuous annular member embedded in the material forming the central region; the end region of the sheath is an end region of a pressure sheath of the flexible pipe defining an internal passage for letting flow a fluid, an end region of an outer sheath of the flexible pipe defining an outer surface of the flexible pipe or/and an end region of an intermediate sheath located in an annular space between the pressure sheath and the outer sheath.

(23) A further subject matter of the invention relates to a flexible pipe comprising a central section including at least one polymer sheath and armor elements arranged around or inside the polymer sheath, the flexible pipe comprising at least one end-fitting as defined hereinabove, mounted at one end of the central section.

(24) A further subject matter of the invention is a method for assembling a flexible pipe, including the following steps: providing an end region of a polymer sheath of the flexible pipe and end

sections of armor elements of the flexible pipe arranged around or in the end region of the polymer sheath; fitting an end vault defining a central bore for the letting the fluid flow along a central axis, and a cover defining, along with the end vault, a chamber for receiving the end sections of armor elements; mounting at least one sealing assembly comprising at least one crimping ring comprising at least one bulge inserted radially into the polymer sheath; placing an annular seal in contact with the polymer sheath away from the crimping ring; characterized in that the annular seal includes an annular central region deformable radially toward the polymer sheath, and two stiffened annular peripheral regions located on both sides axially with respect to the central region, the method comprising moving the peripheral annular regions toward each other in order to activate a radial deformation of the central region toward the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.

(25) The method according to the invention can comprise one or a plurality of the following feature(s), taken individually or according to any technically possible combination: the end-fitting defines a housing for receiving the annular seal, the housing being delimited by a transverse activation surface of the annular seal, located on the end vault or on a part fastened with respect to the end vault, the transverse activation surface extending opposite the first peripheral region, the housing being also delimited by a transverse thrust surface of the annular seal, located opposite the transverse activation surface, on an additional compression activation part of the annular seal, extending opposite the second peripheral region, the movement of the peripheral annular regions toward each other comprising the relative movement of the activation transverse surface and the thrust **20** transverse surface relative to each other, so as to switch the annular seal from the rest configuration to the active configuration thereof; the method comprises the transmission of external pressure to the flexible pipe up to a gap between the activation transverse surface and the first peripheral region, so as to bring the first peripheral region closer to the second peripheral region.

Description

DESCRIPTION OF THE DRAWINGS

(1) The invention will be better understood upon reading the following description, given only as an example and making reference to the enclosed drawings, wherein:

(2) FIG. **1** is a partially exploded perspective view of a first flexible pipe according to the invention;

(3) FIG. **2** is a sectional view along a median axial plane of the relevant parts of the end-fitting of the flexible pipe shown in FIG. **1** illustrating an annular seal applied to the pressure sheath;

(4) FIG. **3** is a view similar to FIG. **2**, in the case of an annular seal applied to an outer sheath;

(5) FIG. **4** is a view similar to FIG. **2**, in the case of an annular seal applied to an intermediate sheath;

(6) FIG. **5** illustrates the principle of operation of the annular seal shown in FIGS. **2** to **4**;

(7) FIG. **6** illustrates a variant of an annular seal that can be used in an end-fitting according to the invention.

DETAILED DESCRIPTION

(8) Hereinafter, the terms “outer” and “inner” generally refer to a radial direction to an axis A-A' of the pipe, the term “outer” meaning relatively further radially from the axis A-A' and the term “inner” meaning relatively closer radially to the axis A-A' of the pipe.

(9) The terms “front” and “rear” refer to an axial direction with respect to an axis A-A' of the pipe, the term “front” meaning relatively further from the middle of the pipe and closer to one end of the pipe, the term “rear” meaning relatively closer to the middle of the pipe and farther from one end of the pipe. The middle of the pipe is the point of the pipe which is equidistant from both ends of the pipe.

- (10) A flexible pipe **10** according to the invention is partially illustrated in FIGS. **1** to **3**. The flexible pipe **10** includes a central section **12** illustrated, in part, in FIG. **1**. The pipe includes, at each of the axial ends of the central section **12**, an end end-fitting **14** (not visible in FIG. **1**), the relevant parts of which are shown in FIGS. **2** and **3**.
- (11) With reference to FIG. **1**, the pipe **10** delimits a central passage **16** for letting a fluid flow, advantageously a petroleum fluid. The central passage **16** extends along a central axis A-A', between the upstream end and the downstream end of the pipe **10**. It emerges through the end-fittings **14**.
- (12) The flexible pipe **10** is intended for being arranged through a body of water (not shown) in an installation for extracting fluids, in particular hydrocarbons.
- (13) The body of water is e.g. a sea, a lake or a river. The depth of the body of water at the installation for using fluids is comprised e.g. between 500 m and 3000 m.
- (14) The installation for using fluids includes a surface unit and a bottom unit (not shown) which are generally connected to each other by the flexible pipe **10**.
- (15) The flexible pipe **10** is preferentially an “unbonded” pipe.
- (16) At least two adjacent layers of the flexible pipe **10** are free to move longitudinally with respect to each other during a bending of the pipe. Advantageously, all the layers of the flexible pipe are free to move with respect to each other. Such a pipe is described e.g. in the normative documents published by the American Petroleum Institute (API), API 17J, 4th edition—May 2014 and API RP 17B, 5th edition—May 2014.
- (17) As illustrated in FIG. **1**, the pipe **10** delimits a plurality of concentric layers about the axis A-A', which extend continuously along the central section **12** as far as the end-fittings **14** situated at the ends of the pipe.
- (18) According to the invention, the pipe **10** includes at least one first polymer sheath advantageously forming a pressure sheath **20**.
- (19) The pipe **10** further includes tensile armor layers **24**, **25** arranged externally with respect to the pressure sheath **20**.
- (20) Advantageously, and depending on the desired use, the pipe **10** further includes an inner carcass **26** arranged inside the pressure sheath **20**, a pressure vault **28** interposed between the pressure sheath **20**, if appropriate a hoop, the layer or layers of tensile armor **24**, **25** and an outer sheath **30**, intended to protect the pipe **10**.
- (21) In a known manner, the tubular sheath **20** is intended to confine, in a leak-tight way, the transported fluid transported, in the passage **16**. It is made of a polymer material, e.g. containing a polyolefin such as polyethylene, containing a polyamide such as PA11 or PA12, or containing a fluoropolymer such as polyvinylidene fluoride (PVDF).
- (22) In a variant, the tubular sheath **20** is made of a high-performance polymer such as PEK (polyether ketone), PEEK (polyether ether ketone), PEEKK (polyether ether ketone), PEKK (polyether ketone ketone), PEKEKK (polyether ketone ether ketone ketone), PAI (polyamide-imide), PEI (polyether-imide), PSU (polysulfone), PPSU (polyphenylsulfone), PES (polyethersulfone), PAS (polyarylsulfone), PPE (polyphenylene ether), PPS (polyphenylene sulfide), LCP (liquid crystal polymers), PPA (polyphthalamide) and/or mixtures thereof or further a mixture with PTFE (polytetrafluoroethylene) or PFPE (perfluoropolyether).
- (23) The thickness of the pressure sheath **20** is comprised e.g. between 5 mm and 20 mm.
- (24) As can be seen in FIG. **2**, the pressure sheath **20** has an end region **27** arranged in the end-fitting **14**.
- (25) The carcass **26**, when present, is formed e.g. by a profiled metal strip, wound in a spiral. The turns of the strip are advantageously interlocked, which serves to take up the radial crushing forces.
- (26) In the present example, the carcass **26** is arranged inside the pressure sheath **20**. The flexible pipe **10** is then referred to by the term “rough bore” because of the geometry of the carcass **26**.
- (27) In a variant (not shown), the flexible pipe **10** does not have an inner carcass **26** and is then

referred to by the term “smooth bore”.

(28) The helical winding of the profiled metal strip forming the carcass **26** has a short pitch, i.e. has a helix angle about the axis A-A' of the pipe of an absolute value close to 90° , typically comprised between 75° and 90° .

(29) In the present example, the pressure vault **28** is intended to take up the radial forces associated with the pressure inside the pressure sheath **20**. The pressure vault is formed e.g. of a profiled metal wire wound helically around the sheath **20**. The profiled wire generally has a complex geometry, in particular in the shape of a Z, a T, a U, a K, an X or an I, which is used to staple the turns of the pressure vault **28** between each other. The stapling of the turns of the pressure vault **28** serves to control the spacings between adjacent turns in order, in particular, to prevent the pressure sheath **20** from creeping through the pressure vault **28** under the effect of the pressure inside the pipe **10**.

(30) The pressure vault **28** is wound in a short-pitch helix around the pressure sheath **20**, i.e. with a helix angle about the axis A-A' of the pipe of an absolute value close to 90° , typically comprised between 75° and 90° .

(31) If appropriate, a hoop (not shown in FIG. 1) surrounds the pressure vault **28**. The hoop is also intended to take up the radial forces associated with the pressure, in addition to the pressure vault **28**. The hoop is formed e.g. of a metal wire of substantially rectangular cross-section wound in a short-pitch helix around the pressure vault sheath **28**. The turns of the hoop are not stapled together.

(32) The flexible pipe **10** according to the invention comprises at least one armor layer **24**, **25** formed by a helical winding of at least one elongated armor element **29**.

(33) In the example shown in FIG. 1, the flexible pipe **10** includes a plurality of armor layers **24**, **25**, in particular a first inner armor layer **24**, applied to the pressure vault **28** (or to the sheath **20** when the vault **28** is absent) and a second outer armor layer **25** around which the outer sheath **30** is arranged.

(34) Each armor layer **24**, **25** includes longitudinal armor elements **29** wound with a large pitch about the axis A-A' of the pipe.

(35) “Wound with a large pitch” means that the absolute value of the helix angle is less than 60° and is typically comprised between 25° and 55° .

(36) The armor elements **29** of a first layer **24** are wound generally following an opposite angle with respect to the armor elements **29** of a second layer **25**. Thereby, if the angle of winding of the armor elements **29** of the first layer **24** is equal to $+\alpha$, α being comprised between 25° and 55° , the angle of winding of the armor elements **29** of the second armor layer **25** arranged in contact with the first layer **24** is e.g. $-\alpha$, with α comprised between 25° and 55° .

(37) The armor elements **29** are e.g. formed by metal wires, or by composite tapes, in particular composite tapes reinforced with carbon fibers.

(38) As can be seen in FIG. 2, the armor elements **29** each have an end portion **32** inserted into the end-fitting **14**. The end portion **32** extends as far as a free end arranged in the end-fitting **14**. It advantageously has a helical or pseudo-helical trajectory of axis A-A' in the end-fitting **14**.

(39) The flexible pipe **10** advantageously includes anti-wear layers interposed between the pressure vault **28** and the first inner armor layer **24**, and between the two armor layers **24**, **25**.

(40) Each anti-wear layer is formed by helically winding a polymeric strip with a thickness typically comprised between 2 mm and 4 mm. Each anti-wear layer has the function of reducing the friction between the metal wires or the wear between the composite tapes between which the layer is inserted.

(41) The outer sheath **30** is intended for forming a barrier sealed against liquids, from the outside of the flexible pipe towards the inside. It is advantageously made of a polymer material, in particular containing a polyolefin, such as polyethylene, containing a polyamide, such as PA11 or PA12, or containing a fluoropolymer such as polyvinylidene fluoride (PVDF).

(42) In a variant, the tubular sheath **30** is made of a high-performance polymer such as PEK (polyether ketone), PEEK (polyether ether ketone), PEEKK (polyether ether ketone), PEKK

(polyether ether ketone ketone), PEKEKK (polyether ketone ether ketone ketone), PAI (polyamide-imide), PEI (polyether-imide), PSU (polysulfone), PPSU (polyphenylsulfone), PES (polyethersulfone), PAS (polyarylsulfone), PPE (polyphenylene ether), PPS (polyphenylene sulfide), LCP (liquid crystal polymers), PPA (polyphthalamide) and/or mixtures thereof or further a mixture with PTFE (polytetrafluoroethylene) or PFPE (perfluoropolyether).

(43) The width of the outer sheath **30** is comprised e.g. between 5 mm and 15 mm.

(44) As illustrated in FIG. 2, each end-fitting **14** includes, in addition to the end region **27** of the pressure sheath **20** and the end sections **32** of the armor elements **29**, an end vault **50** and an outer connection cover **51** protruding axially rearwards from the end vault **50**. The cover **51** delimits, along with the end vault **50**, a chamber **52** for receiving the end sections **32** of the armor elements **29**.

(45) The end-fitting **14** further includes a front sealing assembly **54** around the pressure sheath **20**, and a rear sealing assembly **56** around the outer sheath **30** (visible only in FIG. 3 which will be described hereinbelow).

(46) In the present example, the end-fitting **14** further includes a solid filling material **58**, such as a thermosetting polymer resin such as an epoxy or an araldite. The solid filling material is arranged inside the chamber **52** around the end sections **32**.

(47) In the present example, the end vault **50** is intended to connect the pipe **10** to another connection end-fitting **14** or to end equipment, advantageously by means of an end flange (not shown).

(48) The end vault **50** has a central bore **62** intended to receive the end region **27** of the pressure sheath **20** and to allow the fluid circulating through the central passage **16** to flow to the outside of the pipe **10**.

(49) In the example shown in FIG. 2, the end vault **50** defines internally a rear face **64** delimiting a convergent bearing surface **66** for a crimping ring.

(50) The cover **51** includes a tubular peripheral wall **70** extending around the axis A-A'. The peripheral wall **70** has a front edge (not visible) fastened to the end vault **50**, radially spaced from the armor layers **24**, **25** and a rear edge **74** (visible only in FIG. 3) extending axially rearward beyond the end vault **50**.

(51) The cover **51** delimits the chamber **52** radially outwards.

(52) The front sealing assembly **54** is advantageously located at the front of the end-fitting **14**, in contact with the end vault **50**, and is offset axially forwards with respect to the rear sealing assembly **56**.

(53) It comprises a front crimping ring **76**, intended to engage with the pressure sheath **20**, and a front clamping flange **78** for clamping the front crimping ring **76**.

(54) In the example shown in FIG. 2, wherein the pipe **10** includes a pressure vault **28**, the front sealing assembly **54** further includes a flange **80** for stopping the pressure vault **28**, and advantageously includes a spacer **79**.

(55) According to the invention, the front sealing assembly **54** further comprises an annular seal **81** which can be activated, arranged so as to bear onto the pressure sheath **20**, the annular seal **81** being received in an annular housing **82** of the end-fitting **14**.

(56) The front clamping flange **78** is configured to be screwed onto the rear face **64** of the end vault **50**.

(57) The front crimping ring **76** is configured to be moved forwards along the axis A-A' by sliding on the convergent surface **66**, being thrust forwards by the front clamping flange **78**, during the mounting thereof on the end vault **50**. The front crimping ring **76** includes a front bulge **84** configured to be thrust radially into the pressure sheath **20** by a wedge effect during the movement of the front crimping ring **76** on the convergent surface **66**.

(58) As can be seen in FIGS. 2 and 5, the housing **82** is defined at the front by an annular shoulder formed in the flange **78**, facing the end region **27** of the pressure sheath **20** arranged in the end-

fitting. The shoulder defines a transverse surface **90** for activating the annular seal **81**, arranged in front of the annular seal **81**, and an inner circumferential face **91** extending around and facing the sheath **20**.

(59) The housing **82** is closed toward the rear by a transverse thrust surface **92** defined at the front of the spacer **79**. Thereby, the transverse thrust surface **92** is configured to be brought closer to the transverse activation surface **90** in order to compress the annular seal **81**. The spacer **79** forms herein an additional part for activating the compression of the annular seal **81**, directly mounted onto the end vault **50**.

(60) With reference to FIG. 5, the end vault **50** and the front flange **78** advantageously define at least one pressure transmission channel **94** external to the flexible pipe **10** toward the housing **82**.

(61) The channel **94** opens out, on the one hand, through the transverse activation surface **90**, and, on the other hand, into the chamber **52**. Thereby, the external pressure which is applied, if appropriate, in the chamber **52**, is configured to be transmitted to the intermediate space situated between the transverse activation surface **90** and the annular seal **81** through the channel **94**.

(62) As illustrated in FIGS. 2 and 5, the annular seal **81** which can be activated, includes a deformable central region **100**, intended to be inserted in a sealed way between the sheath **20** and the inner face **91** delimiting the housing **82**, a first stiffened front peripheral region **102**, arranged on a first side of the central region **100** and a second stiffened rear peripheral region **104** arranged on another side of the central region **100**. The first peripheral region **102** and the second peripheral region **104** are movable toward each other to compress the central region **100** and cause the radial expansion of the central region **100** toward the sheath **20**.

(63) The central region **100** is preferentially formed of a deformable material, such as an elastomer. The elastomer is e.g. a fluoroelastomer, such as fluorocarbon (FKM) manufactured with vinylidene fluoride (VDF or VF2) chosen e.g. amongst a hexafluoropolypropylene (HFP) and VF2 copolymer, a tetrafluoroethylene (or "TFE"), HFP, and VF2 terpolymer, a TFE and perfluoromethylvinylether (or "PMVE") and VF2 terpolymer, a TFE, propylene and VF2 terpolymer or a TFE, HFP, ethylene, PMVE and VF2 pentapolymer.

(64) In a variant, the fluoroelastomer is a perfluoroelastomer (FFKM), e.g. a terpolymer of TFE, of PMVE and of a third monomer which can be used in crosslinking. In another variant, the fluoroelastomer is a fluorosilicone (FVMQ) or a propylene tetrafluoroethylene (FEPM) copolymer.

(65) In a variant or in addition, the central region **100** is made of a nitrile rubber, more particularly an acrylonitrile butadiene copolymer (NBR) and/or a hydrogenated acrylonitrile butadiene copolymer (HNBR).

(66) In the present example, the central region **100** has a trapezoidal cross-section tapering from the outside toward the inside between the inner face **91** of the housing **82** and the pressure sheath **20**, the cross-section being taken in each median axial plane passing through the axis A-A'.

(67) With reference to FIG. 5, the central region **100** comprises a first inclined front surface **106**, a second inclined rear surface **108**, and an outer circumferential surface **110** connecting the first inclined surfaces **106** and **108**.

(68) It further includes an inner tip **112** intended to be applied on the sheath **20**.

(69) The first surface **106** is inclined backwards, moving from the front to the rear. It converges toward the inner tip **112**. The second inclined surface **108** is inclined forwards by moving from the rear to the front. It also converges toward the tip **112**.

(70) Thereby, the width of the central region **100** taken along the axis A-A' at the inner tip **112** is less than the width of the central region **100** taken along the axis A-A' at the outer circumferential surface **110**.

(71) Preferentially, the hardness of the material forming the central region **100** is less than 100 Shore A, in particular between 60 Shore A and 90 Shore A.

(72) The central region **100** is thereby more deformable than the peripheral regions **102**, **104**, allowing the radial expansion thereof toward the sheath **20** and the compression thereof on the

sheath **20**.

(73) The first front peripheral region **102** has a wedge-shaped section taken in each median axial plane passing through the axis A-A'. It thereby has a first front lateral surface **114**, perpendicular to the axis A-A', intended to extend opposite the transverse activation surface **90** and a first supplementary inclined surface **116**, situated at the rear, applied to the first front inclined surface **106** of the central region **100**.

(74) The angle α of inclination of the first inclined front surface **106** and of the first supplementary inclined surface **116** is comprised e.g. between 40° and 80° with respect to the axis A-A'.

(75) The first front lateral surface **114** is advantageously continued toward the sheath **20** by a front chamfer **114A** situated opposite the sheath **20**.

(76) The second peripheral region **104** has a second rear lateral surface **118**, perpendicular to the axis A-A', intended to extend opposite the transverse thrust surface **92**, and a second supplementary inclined surface **120**, situated at the front, applied against the second rear inclined surface **108** of the central region **100**.

(77) The angle β of inclination of the second rear inclined surface **108** and of the second supplementary inclined surface **116** with respect to the axis A-A' is comprised e.g. between 40° and 80°. The angle β is preferentially equal to the angle α .

(78) The second rear lateral surface **118** is advantageously continued toward the sheath **20** by a rear chamfer **118A** situated opposite the sheath **20**.

(79) The first front peripheral region **102** and the second rear peripheral region **104** are each formed of a material that is stiffer than the central region, e.g. a thermoplastic polymer.

(80) It is made e.g. of PEK (polyetherketone), PEEK (polyetheretherketone), PEEKK (polyetheretherketone ketone), PEKK (polyetherketone ketone), PEKEKK (polyetherketone ether ketone ketone), or of a fluoropolymer, such as polyethylene tetrafluoride (PTFE).

(81) Preferentially, the hardness of the material forming each peripheral region **102**, **104** is greater than the hardness of the central region **100**. The hardness is e.g. greater than 65 shore D.

(82) The peripheral region **102**, **104** is e.g. made of PTFE or of PEEK.

(83) Thereby, under the effect of a rearward thrust force applied to the first peripheral region **102** at the first front lateral surface **114** and/or a forward thrust force applied to the second peripheral region **104** at the second lateral surface **118**, the first peripheral region **102** and the second peripheral region **104** are configured to approach each other to compress the central region **100** between the supplementary inclined surfaces **116**, **120**. The above causes the radial expansion of the inner tip **112** toward the sheath **20**, and the application of the circumferential surface **110** on the inner face **91** of the housing **82**.

(84) The annular seal **81** thereby switches from a rest configuration to a first active sealing configuration, in compression, wherein said seal forms a sealing between the sheath **20** and the front clamping flange **78** during the assembly of the end-fitting **14**.

(85) Such sealing is reinforced when the pressure increases in the annulus between the outer sheath **30** and the pressure sheath **20**, tending to thrust the second peripheral region **104** toward the first peripheral region **102**.

(86) In a known way, as illustrated in FIG. 3, the rear sealing assembly **56** includes a rear crimping ring **130**, a rear clamping flange **132** for clamping the rear crimping ring **130**, and a rear compression flange **134**.

(87) The rear sealing assembly **56** advantageously includes an intermediate support sleeve **128** interposed between the outer sheath **30** and the armor layers **24**, **25**, the outer sheath **30** resting on the intermediate sleeve **128**.

(88) Advantageously, in the embodiment shown in FIG. 3, the rear sealing assembly **56** further includes an external annular seal **81** which can be activated, arranged in an outer housing **82**.

(89) The rear crimping ring **130** is interposed between a rear inclined surface **136** of the cover **51** and the outer sheath **30**. It has a front bulge **137** configured to be radially thrust into the outer

sheath **30** by a wedging effect during the movement of the rear crimping ring **130** on the surface **136**.

(90) The rear clamping flange **132** is fastened on the rear edge **74** of the cover **51**. It thrusts forward the rear crimping ring **130** to move the ring radially toward the axis A-A' and crimp the outer sheath **30**.

(91) The outer sheath **30** bears on the intermediate sleeve **128**. It is axially wedged to the rear of the fastening assembly **56**.

(92) The outer housing **82** of the rear sealing assembly **56** has a similar structure to the housing **82** defined for the front sealing assembly **54**.

(93) It is defined by an annular shoulder formed in the rear clamping flange **132**, facing an end region of the outer sheath **30** arranged in the end-fitting **14**. As hereinabove, the shoulder defines a first transverse surface **90** for activating the annular seal **81**, arranged in front of the annular seal **81** and an inner circumferential face **91** extending around and facing the sheath **30**.

(94) The housing **82** is closed toward the rear by a transverse thrust surface **92** defined at the front of the rear compression flange **134**. Thereby, the transverse thrust surface **92** is configured to be brought closer to the transverse activation surface **90** in order to compress the annular seal **81** as described hereinabove. The rear compression flange **134** thereby forms an additional part for activating the compression of the annular seal **81**, directly mounted onto the cover **51**.

(95) The annular seal **81** of the rear seal assembly **56** has an identical structure to the annular seal **81** of the front seal assembly **54**.

(96) The assembly of the end-fitting **14** will now be described.

(97) Initially, the end vault **50** is fitted around the end region **27** of the pressure sheath **20**, the armor layers **24**, **25** having been moved radially outwards.

(98) The crimping ring **76** is fitted between the pressure sheath **20** and the convergent surface **66** of the end vault **50**.

(99) The clamping flange **78** is then fitted to be clamped against the rear face **64** of the end vault **50**. Then, the annular seal **81** which can be activated in the rest configuration thereof, is fitted in the housing **82**. The housing **82** is then closed by inserting the spacer **79**.

(100) The clamping flange **78** and the spacer **79** are then clamped against the end vault **50**, making the transverse thrust surface **92** move toward the transverse activation surface **90**, and the central region **100** of the joint which can be activated, compress between the peripheral regions **102**, **104**. As a result, the inner tip **112** is applied and compressed against the sheath **20**.

(101) Similarly, the sleeve **128** is inserted between the armor elements **24**, **25** and the outer sheath **30**.

(102) The cover **51** is fitted around the pipe **10** so as to form, with the end vault **50**, the chamber **52**. The rear crimping ring **130** is then fitted between the cover **51** and the outer sheath **30**. The rear clamping flange **132** is then fitted so as to delimit the outer housing **82**, then the annular seal **81** is inserted into the outer housing **82**.

(103) The compression flange **134** is then assembled at the rear of the clamping flange **132**. The rear clamping flange **132** is then clamped against the rear edge **74** of the cover **51**. The compression flange **134** moves closer to the rear clamping flange **132**, causing the compression of the central region **100** of the outer annular seal **81** between the first peripheral region **102** and the second peripheral region **104**, and the application of the inner end-fitting **112** against the outer sheath **30**.

(104) Advantageously, during assembly, the sealing provided by the annular seal **81** can be tested by inserting a pressure through the channel **94**, following an adaptation of the method described in U.S. Pat. No. 6,923,477, in order to ensure that, in the event of a failure of the sealing between the front crimping ring **76** and the pressure sheath **20**, the annular seal **81** is configured to provide the sealing between the end vault **50** and the pressure sheath **20**.

(105) In use, in the event of a possible overpressure on the outside of the flexible pipe **10** with respect to the inside of the flexible pipe **10**, the annular seal **81** maintains the sealing around the

pressure sheath **20** even if the integrity of the crimping front ring **76** with respect to the end region **27** of the pressure sheath **20** fails. The sealing is even reinforced, since the excess pressure transmitted to the annular seal **81** through the channel **94** causes an additional compression of the central region **100** beyond the primary compression resulting from the fitting of the annular seal **81**.

(106) In the variant shown in FIG. **4**, the flexible pipe **10** further comprises an intermediate sheath **150** arranged in the annulus between the pressure sheath **20** and the outer sheath **30**. The end-fitting further comprises an intermediate sleeve **152** on which the intermediate sheath **150** rests.

(107) The intermediate sleeve **152** is advantageously provided with an internal passage for gas circulation.

(108) The end-fitting **14** comprises an intermediate sealing assembly **153** comprising a first intermediate housing flange **154** mounted on the end vault **50** at the rear face **64** and a second intermediate compression flange **156** mounted on the first intermediate flange **154**. The intermediate sealing assembly **153** further comprises an intermediate ring **158** for crimping the intermediate sheath **150** and a clamping flange **159** mounted at the rear of the flanges **154**, **156**.

(109) Advantageously, in the embodiment shown in FIG. **4**, the intermediate sealing assembly **153** further includes an intermediate annular seal **81** which can be activated, arranged in an intermediate housing **82** formed between the first intermediate flange **154** and the second intermediate flange **156**.

(110) The intermediate crimping ring **158** includes two opposite bulges, a front bulge **160** and a rear bulge **162**, each intended to be inserted into the intermediate sheath **150**.

(111) The intermediate crimping ring **158** is interposed at the front, between a rear inclined surface **164** of the intermediate crimping flange **156** and the intermediate sheath **150**, and at the rear, between a front inclined surface **166** of the intermediate clamping flange **159** and the intermediate sheath **150**.

(112) The above causes the crimping of the front bulge **160** and of the rear bulge **162** into the intermediate sheath **150**.

(113) The intermediate housing flange **154**, the intermediate crimping flange **156** and the intermediate clamping flange **159** are pressed against each other and are fastened together to the rear face **64** of the end vault **50**.

(114) The intermediate sheath **150** bears on the intermediate sleeve **152**.

(115) The intermediate housing **82** of the intermediate sealing assembly **153** has a similar structure to the housing **82** defined for the front sealing assembly **54**. It is located at the front of the intermediate crimping ring **158**.

(116) The intermediate housing **82** is defined by an annular shoulder formed in the intermediate housing flange **154**, facing an end region of the intermediate sheath **150** arranged in the end-fitting **14**. As hereinabove, the shoulder defines a first transverse surface **90** for activating the annular seal **81**, arranged in front of the annular seal **81** and an inner circumferential face **91** extending around and facing the intermediate sheath **150**.

(117) The housing **82** is closed toward the rear by a transverse thrust surface **92** defined at the front of the intermediate compression flange **156**. Thereby, the transverse thrust surface **92** is configured to be brought closer to the transverse activation surface **90** in order to compress the annular seal **81**, as described hereinabove.

(118) The intermediate compression flange **156** thereby forms an additional part for activating the compression of the annular seal **81**, directly mounted on the end vault **50**.

(119) The annular seal **81** of the intermediate seal assembly **153** has an identical structure to the annular seal **81** of the front seal assembly **54**. It provides an effective seal between the intermediate sheath **150** and the intermediate flanges **154**, **156**, even if the seal is not maintained by the intermediate crimping ring **158**.

(120) In a variant applying to each of the embodiments shown in FIGS. **2** to **4**, the first peripheral region **102** and the second peripheral region **104** are each formed by a stiffened ring **180** embedded

in the deformable material forming the central region **100**. The stiffened ring **180** is e.g. a spring wire, a rod or a rope.

(121) The stiffened ring **180** is e.g. formed of metal, a metal alloy or a material that is stiffer than the central region **100**, e.g. a thermoplastic polymer, in particular fibers.

(122) The metal alloy is e.g. phosphor bronze. The fibers are e.g. synthetic polyester fibers.

(123) The stiffer material is e.g. made of PEK (polyetherketone), PEEK (polyetheretherketone), PEEKK (polyetheretherketoneketone), PEKK (polyetherketone), PEKEKK (polyetherketonetherketone ketone), or a fluoropolymer, such as polyethylene tetrafluoride (PTFE). The thermoplastic polymer may be filled with fibers such as carbon fibers.

(124) In a variant, the stiffened ring **180** is made of a combination of metal and thermoplastic polymer, e.g. arranged coaxially. More particularly, the stiffened ring **180** comprises coaxial coil springs, advantageously a spring with an internal coil made of metal, in particular stainless steel, and an external spring made of polymer, in particular PEEK (polyetheretherketone).

(125) In a variant, as illustrated in FIG. 6, the stiffened ring **180** is a combination of a spring wire **180A** and a snap ring **180B**, the snap ring **180B** being arranged inside the spring wire **180A**.

(126) In the present example, the stiffened ring **180** is arranged in the vicinity of an inner surface of the annular seal **81** intended to be applied on the sheath **20, 30, 150** on both sides of the end-fitting **112**. Herein, at rest, the tip **112** protrudes. The tip has a rounded contour.

(127) Moreover, the operation of the annular seal **81** described in FIG. 6 is analogous to the operation described in FIG. 5.

Claims

1. An end-fitting of a flexible pipe for transporting a fluid, comprising: at least one end region of a polymer sheath of the flexible pipe; end sections of armor elements of the flexible pipe arranged around or in the end region of the polymer sheath; an end vault defining a central bore for letting the fluid flow, extending along a central axis; a cover defining, along with the end vault, a chamber for receiving the end sections of armor elements; at least one sealing assembly comprising at least one crimping ring, the crimping ring comprising at least one bulge radially inserted into the polymer sheath; an annular seal, arranged in contact with the polymer sheath away from the crimping ring; the annular seal including a central annular region deformable radially toward the polymer sheath, and two stiffened annular peripheral regions, located on either side of the central region along the central axis, the peripheral annular regions being movable toward each other to activate the radial deformation of the central region toward the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.
2. The end-fitting according to claim 1, wherein at least in the operative position, the central region defines a radial tip protruding toward the polymer sheath beyond each peripheral annular region.
3. The end-fitting according to claim 2, wherein in the rest configuration, the central region has the radial tip.
4. The end-fitting according to claim 1, wherein the central region (**100**) is made of elastomer, each peripheral region (**102, 104**) being made of metal, or of a polymer that is more rigid than the elastomer.
5. The end-fitting according to claim 4, wherein the elastomer is a fluoroelastomer or a nitrile elastomer.
6. The end-fitting according to claim 4, wherein the polymer which is more rigid than the elastomer, contains PEK, PEEK, PEEKK, PEKEKK, or contains a fluoropolymer.
7. The end-fitting according to claim 1, defining a housing for receiving the annular seal, the housing being delimited by a transverse activation surface of the annular seal, located on the end vault or on a part fastened with respect to the end vault, the transverse activation surface extending opposite the first peripheral region, the housing being also delimited by a transverse thrust surface

of the annular seal, located opposite the transverse activation surface, on an additional compression activation part of the annular seal, the transverse thrust surface extending opposite the second peripheral region, the transverse activation surface and the transverse thrust surface being movable relative to each other at least during the fitting of the end-fitting so as to switch the annular seal from the rest configuration of the annular seal to the active configuration of the annular seal.

8. The end-fitting according to claim 7, wherein the additional part is fastened onto the crimping ring.

9. The end-fitting according to claim 7, defining a channel for transmitting an external pressure to the flexible pipe, the channel opening out into the housing between the transverse activation surface and the first peripheral region.

10. The end-fitting according to claim 1, wherein the first peripheral region and/or the second peripheral region are directly mounted on surfaces of the central region.

11. The end-fitting according to claim 10, wherein the central region has a first inclined surface at a non-zero angle with respect to a plane perpendicular to the central axis, the first peripheral region having a first supplementary inclined surface, directly mounted onto the first inclined surface, the inclination of the first inclined surface being directed toward the second peripheral region when moving radially towards the central axis on the first inclined surface.

12. The end-fitting according to claim 11, wherein the central region has a second inclined surface at a non-zero angle with respect to a plane perpendicular to the central axis the second inclined surface being located opposite the first inclined surface along the central axis, the second peripheral region having a supplementary second inclined surface fastened to the second inclined surface.

13. The end-fitting according to claim 12, wherein the central region has a trapezoidal section.

14. The end-fitting according to claim 1, wherein the first peripheral region and the second peripheral region each have a continuous or discontinuous annular member embedded in the material forming the central region.

15. The end-fitting according to claim 1, wherein the end region of the sheath is an end region of a pressure sheath of the flexible pipe defining an internal passage for letting flow a fluid, an end region of an outer sheath of the flexible pipe defining an outer surface of the flexible pipe and/or an end region of an intermediate sheath located in an annular space between the pressure sheath and the outer sheath.

16. A flexible pipe, comprising a central section including a polymer sheath and armor elements arranged around or inside the polymer sheath, the flexible pipe comprising at least one end-fitting mounted at one end of the central section, the at least one end fitting including: at least one end region of the polymer sheath; end sections of the armor elements arranged around or in the end region of the polymer sheath; an end vault defining a central bore for letting the fluid flow, extending along a central axis; a cover defining, along with the end vault, a chamber for receiving the end sections of armor elements; at least one sealing assembly comprising at least one crimping ring, the crimping ring comprising at least one bulge radially inserted into the polymer sheath; an annular seal, arranged in contact with the polymer sheath away from the crimping ring; the annular seal including a central annular region deformable radially toward the polymer sheath, and two stiffened annular peripheral regions, located on either side of the central region along the central axis, the peripheral annular regions being movable toward each other to activate the radial deformation of the central region toward the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.

17. A method of assembly of an end-fitting of a flexible pipe, comprising: providing an end region of a polymer sheath of the flexible pipe and end sections of armor elements of the flexible pipe arranged around or in the end region of the polymer sheath; fitting an end vault defining a central bore for the letting the fluid flow along a central axis, and a cover defining, along with the end vault, a chamber for receiving the end sections of armor elements; assembling at least one sealing assembly comprising at least one crimping ring, the crimping ring comprising at least one bulge

inserted radially into the polymer sheath; placing an annular seal in contact with the polymer sheath away from the crimping ring; the annular seal including a central annular region deformable radially toward the polymer sheath, and two stiffened annular peripheral regions located axially on opposite sides of the central region, the method comprising moving the peripheral annular regions toward each other to activate a radial deformation of the central region toward the polymer sheath from a rest configuration to an active sealing configuration around the polymer sheath.

18. The method according to claim 17, wherein the end-fitting defines a housing for receiving the annular seal, the housing being delimited by a transverse activation surface of the annular seal, located on the end vault or on a part fastened with respect to the end vault, the transverse activation surface extending opposite the first peripheral region, the housing being also delimited by a transverse thrust surface of the annular seal, located opposite the transverse activation surface, on an additional compression activation part of the annular seal, extending opposite the second peripheral region, the movement of the peripheral annular regions toward each other comprising the relative movement of the activation transverse surface and the thrust transverse surface relative to each other, so as to switch the annular seal from the rest configuration to the active configuration of the annular seal.

19. The method according to claim 18, comprising the transmission of an external pressure to the flexible pipe as far as a gap between the activation transverse surface and the first peripheral region to bring even closer the first peripheral region to the second peripheral region.
